

Chapter 1: Project Overview

1.1 Agriculture in India:

India, renowned for its rich agricultural heritage, stands as a testament to the symbiotic relationship between its people and the land they cultivate. With diverse terrain and varied climatic conditions, agriculture has been both a cornerstone of the economy and a cultural touchstone for centuries. From the fertile plains of Punjab to the terraced fields of the Himalayas and the sun-kissed orchards of Tamil Nadu, Indian agriculture mirrors the nation's diversity. Yet, despite its significance, the sector faces numerous challenges, ranging from unpredictable weather patterns to diminishing arable land & an ever-growing demand for food.

1.2 The Emergence of Smart Agro Bots:

The convergence of agriculture and robotics has given rise to the concept of smart agro bots – intelligent robotic systems specifically designed to enhance agricultural productivity, sustainability, and resilience. Unlike traditional farming machinery, smart agro bots are equipped with advanced sensors, actuators, and artificial intelligence algorithms, enabling them to perceive, analyze, and respond to dynamic environmental conditions in real time.

These smart bots are capable of performing a wide range of tasks with unparalleled precision and efficiency. They can autonomously navigate through fields, identify and selectively remove weeds, monitor crop health, apply fertilizers and pesticides with pinpoint accuracy, and even harvest crops at optimal ripeness. By leveraging data-driven insights and predictive analytics, smart agro bots empower farmers to make informed decisions and optimize resource allocation for maximum yield and profitability. In essence, smart agro bots represent a paradigm shift in agricultural technology, offering a holistic approach to farming that integrates the principles of precision agriculture, automation, and artificial intelligence. By harnessing the power of robotics to augment human labor and intelligence, these innovative solutions hold the key to a more sustainable, resilient, and productive future for Indian agriculture.

As we delve deeper into the realm of smart agro bots, this report aims to explore the various applications, benefits, challenges, and future prospects of this transformative technology in the context of Indian agriculture.

1.3 Introducing Crop Crafter: Modern Ag Revolution

Crop Crafter marks a significant advancement in agricultural technology, providing innovative automation and precision farming techniques to transform traditional farming methods. This groundbreaking system seamlessly integrates advanced features such as sensor data collection, image processing, and robotic arm manipulation. By harnessing cutting-edge technologies like machine learning and ESP32 CAM modules, it empowers farmers to optimize crop cultivation, enhance productivity, and minimize resource wastage. With its ability to perform tasks ranging from soil analysis to crop harvesting with unparalleled accuracy and efficiency, Crop Crafter promises to redefine the landscape of modern agriculture.

1.4 Features/Functionalities:

- Crop Crafter navigates through crops using obstacle avoidance sensors.
- Captures images to identify crop types and assess crop health.
- Measures soil moisture and temperature for irrigation management.
- Adjusts the irrigation system based on collected data for optimal growth.
- Robotic arm harvests fruits, samples soil, and adjusts sensors autonomously

Chapter 2: Methodology

2.1 Project Flow:

Crop Crafter begins with the initialization of the robot, achieved by moving the car to its starting position. Once initialized, the robot collects sensor data to measure various environmental parameters such as soil moisture, temperature, and humidity. This data is then processed and displayed on the accompanying mobile app, providing real-time insights for farmers. Additionally, image processing is performed using an ESP32 CAM module to capture and analyze images of the field. Machine learning algorithms enable object detection within these images, identifying crops, weeds, and other objects of interest. Based on the coordinates of detected objects, the robotic arm moves accordingly, performing tasks such as harvesting, weed removal, or sample collection with precision and efficiency.

The overall flow of the Agri-bot is explained further by a flowchart:

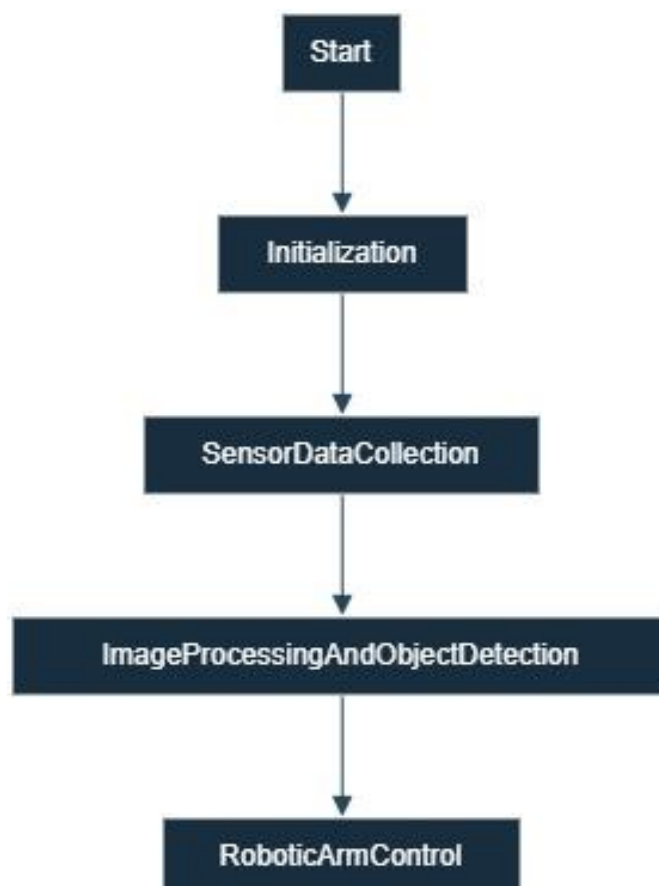


Fig -1: Flowchart

Chapter 3: Preliminary Design

3.1 Hardware Implementation

3.1.1 Servo Motor:

The SG90 servo motor is a compact and cost-effective device known for its widespread use in robotics and electronic projects. Operating on a 5V power supply, it offers a 180-degree rotation span with precise control. Characterized by its lightweight design and affordability, the SG90 servo is favored for its suitability in applications where controlled mechanical movements are essential. Its versatility makes it a popular choice among hobbyists and professionals alike, serving various purposes in the construction of robotic systems, model vehicles, and other projects requiring accurate and controlled motion. The SG90's reliability and ease of use contribute to its widespread adoption in the realm of electronics and automation.



Fig -2: SG 90 Servo Motor

3.1.2 Ultrasonic Sensor:

The ultrasonic sensor is a versatile device employed for distance measurement and object detection. It operates on the principle of emitting ultrasonic waves and measuring their reflection time to determine the distance to an object. Utilizing the echo produced by the waves, it calculates distances accurately, making it integral in robotics, automation, and security systems. Renowned for its non-contact and precise sensing capabilities, the ultrasonic sensor excels in environments where traditional methods may be impractical. Its applications span obstacle avoidance in robotics, parking assistance in vehicles, and fluid level detection.



Fig -3: Ultrasonic Sensor

3.1.3 Temperature Sensor:

The **DHT11** is a commonly used **Temperature and humidity sensor**. The sensor comes with a dedicated NTC to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data. The sensor is also factory-calibrated and hence easy to interface with other microcontrollers. The sensor can measure temperature from 0°C to 50°C and humidity from 20% to 90% with an accuracy of $\pm 1^\circ\text{C}$ and $\pm 1\%$. Operating on a single-wire communication protocol, it provides real-time data output, making it ideal for various applications such as weather stations, HVAC systems, and environmental monitoring devices.

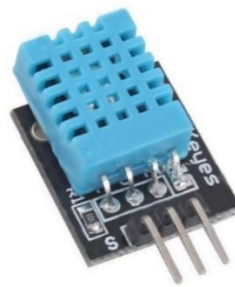


Fig - 4: DHT 11 Temperature Sensor

3.1.4 Moisture Sensor:

The FC-28 soil moisture sensor is a soil hygrometric transducer designed to measure the moisture content in the surrounding soil. Utilizing two probes, it passes current through the soil and reads the resulting resistance to determine the moisture level. Increased moisture enhances soil conductivity, resulting in lower resistance, while dry soil exhibits poor conductivity, leading to higher resistance. This principle allows the sensor to provide accurate readings of soil moisture levels. Widely employed in agriculture, gardening, and research, it enables precise monitoring and control of irrigation systems, promoting optimal plant growth and resource utilization.

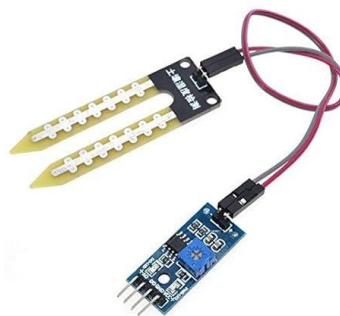


Fig - 5: Soil Moisture Sensor

3.1.5 Motor Driver:

The L298N motor driver is a robust and widely used integrated circuit designed for driving and controlling DC motors. Operating on a dual H-bridge configuration, it can control the direction and speed of two motors independently. Its versatility and high current capability make it a popular choice in robotics and mechatronics projects. The L298N accepts both TTL and CMOS logic inputs, offering flexibility in interfacing with microcontrollers. With built-in protection diodes, it safeguards the circuit from back electromotive force generated by the motors. This motor driver's reliability, ease of use, and ability to handle substantial loads contribute to its prominence in applications requiring precise motor control.

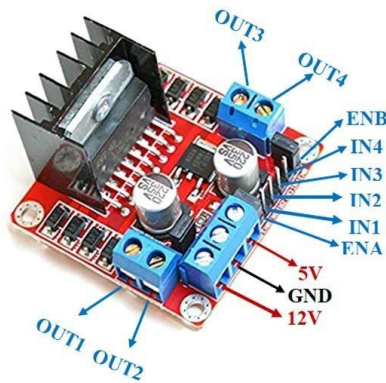


Fig -6: L298DN Motor Driver

S.NO	OPERATION
ENA & ENB	Enables PWM signal for Motor A, Enables PWM signal for Motor B.
12 V	12V input from DC power Source
GND	Ground Pin
OUT1 & OUT 2	Output pins of Motor A
OUT1 & OUT 2	Output pins of Motor B
IN1 & IN2	Motor A input pins. Used to control the spinning direction of Motor A
IN3 & IN4	Motor B input pins. Used to control the spinning direction of Motor B.

Table -1: Logic table of L298N motor driver

3.1.6 Arduino Board:

The Arduino Uno board is a popular microcontroller development board renowned for its versatility and ease of use. It features an Atmega328P microcontroller at its core, offering a wide range of digital and analog input/output pins for connecting sensors, actuators, and other peripherals. Equipped with a USB interface, it allows for easy programming and communication with a computer. The Arduino Uno is compatible with the Arduino IDE, a user-friendly software platform for writing, compiling, and uploading code to the board. It's widely used in hobbyist projects, educational settings, and prototyping applications, enabling enthusiasts and professionals alike to bring their ideas to life through programming and electronics. With its robust design and extensive community support, the Arduino Uno remains a cornerstone in the world of embedded systems development.



Fig - 7: Arduino UNO

3.1.7 Wi-Fi Module:

The WiFi module **NodeMCU ESP8266** is a versatile and widely used microcontroller unit (MCU) that integrates WiFi connectivity. It is based on the ESP8266 chip and features Lua-based firmware, making it a powerful and user-friendly platform for Internet of Things (IoT) projects. The NodeMCU ESP8266 supports seamless communication with WiFi networks, allowing devices to connect and exchange data wirelessly. Its compact form factor and built-in USB-to-serial converter simplify programming and development. The module facilitates real-time data transfer and is extensively utilized in home automation, smart devices, and various IoT applications.



Fig - 8: Wi-Fi Module

3.1.8 WiFi/Bluetooth Module:

The **ESP32-CAM** is a compact yet powerful development board based on the ESP32 microcontroller and featuring an OV2640 camera module. It's designed for applications requiring camera functionality alongside the processing power and connectivity options provided by the ESP32 chip. With onboard Wi-Fi and Bluetooth capabilities, it enables wireless communication and can be utilized in a wide range of IoT projects and applications such as surveillance cameras, video streaming, facial recognition, and more. The ESP32-CAM also supports microSD card storage, allowing for local storage of captured images and videos. Additionally, it can be programmed using the Arduino IDE, making it accessible to a broad community of developers.



Fig - 9: ESP 32 Cam Module

3.1.9 FTDI Programmer:

The FTDI programmer interfaces with the ESP32-CAM module for code upload due to its lack of a built-in USB connector. It connects to the ESP32-CAM's UART pins (U0R and U0T) for serial communication. This setup allows developers to program the ESP32-CAM using the Arduino IDE or similar environments. Additionally, the FTDI programmer provides power and ground connections to the ESP32-CAM. The connection between the FTDI programmer and the ESP32-CAM module typically looks like this:

- FTDI TX (transmit) -> ESP32-CAM U0R (receive)
- FTDI RX (receive) -> ESP32-CAM U0T (transmit)
- FTDI GND (ground) -> ESP32-CAM GND (ground)
- FTDI VCC (power) -> ESP32-CAM 5V or 3.3V



Fig - 10: FTDI Module

3.1.10 Relay Module:

The Single Channel Relay Module is a convenient board that can be used to control high voltage, high current loads such as motors, solenoid valves, lamps, and AC loads. It is designed to interface with microcontrollers such as Arduino, NodeMCU, etc. The relay's terminal (COM, NO, and NC) is being brought out with a screw terminal. It also comes with an LED to indicate the status of the relay. The relay is the device that opens or closes the contacts to switch ON/OFF other appliances operating at high voltages. It is also used in safety circuits where it detects undesirable conditions with an assigned area and gives commands to the circuit breaker to disconnect the affected area through ON or OFF.



Fig -11: Single Channel Relay

3.1.11 DC Motor:

The DC motors are integral components in robotics projects, providing mechanical motion through their rotating shafts. These motors convert electrical energy into mechanical motion, enabling robots to move wheels, limbs, or other components. DC motors offer precise control over speed and direction, crucial for maneuverability in robotics. By adjusting voltage levels or using motor drivers, their rotational characteristics can be finely tuned. Typically powered by batteries, DC motors are reliable and efficient, making them ideal for tasks like propulsion, manipulation, or sensing in diverse robotic applications. Their simplicity, versatility, and controllability contribute to their widespread use, forming the backbone of many robotic systems.



Fig - 12: DC Motor

3.1.12 Solar Panel:

The 4V Solar Cell Panel is a compact and efficient photovoltaic module designed to harness sunlight for electrical power generation. With a voltage rating of 4VDC and an open-circuit voltage of 4.4VDC, it provides a reliable power source for low-power applications. This panel's current rating of 100mA enables it to deliver a steady flow of electrical current under sunlight. Ideal for charging small batteries, powering low-power devices, or integrating into solar-powered systems, it offers versatility in various applications such as remote monitoring, outdoor lighting, and educational projects. Its compact size and efficient performance make it a convenient and sustainable energy solution.

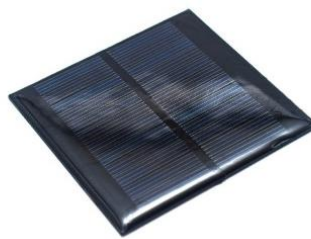


Fig -13: Solar Panel

3.1.13 Rechargeable Battery:

Rechargeable batteries are energy storage devices that can be recharged and reused multiple times, offering a cost-effective and environmentally friendly alternative to single-use disposable batteries. They come in various chemistries such as nickel-cadmium (NiCd), nickel-metal hydride (NiMH), and lithium-ion (Li-ion), each with its characteristics in terms of capacity, voltage, and cycle life. Rechargeable batteries are commonly used in portable electronic devices like smartphones, laptops, cameras, and power tools, as well as in renewable energy systems, electric vehicles, and backup power applications. They help reduce waste and contribute to sustainable energy practices by minimizing the need for single-use batteries.



Fig -14: Rechargeable Battery

3.1.14 Robotic Arm:

The DIY Acrylic Robot Manipulator Mechanical Arm is a versatile and precise robotic arm featuring four degrees of freedom (DOF). Crafted from laser-cut acrylic mechanical parts, it offers exceptional precision and durability, making it ideal for various applications. Its modular design allows for easy assembly and customization to suit specific needs. Primarily utilized for pick-and-place operations, this mechanical arm excels in handling objects with precision and efficiency. Whether in educational settings, research laboratories, or industrial environments, its flexibility and reliability make it a valuable tool for automation tasks. With its intuitive controls and robust construction, users can effortlessly program and operate the arm to perform intricate tasks with ease. From manufacturing processes to educational demonstrations, the Robot Manipulator Mechanical Arm delivers exceptional performance, empowering users to explore the realms of robotics and automation effectively.



Fig -15: Robotic Arm

3.1.15 Water Pump:

A water pump is a mechanical device used to transfer water from one location to another. It operates by creating suction or pressure to move water through pipes or hoses. Water pumps are available in various types, including centrifugal pumps, submersible pumps, and diaphragm pumps, each suited for specific applications such as irrigation, drainage, or water supply systems. They are essential in industries, agriculture, construction, and residential settings, providing reliable water transportation for a wide range of purposes.



Fig -16: Water Pump

3.1.16 Jumper Wires:

Jumper wires are essential electrical components used in electronics and prototyping. These flexible wires, featuring connectors at each end, facilitate seamless connections between various electronic components on a breadboard or circuit. Available in different lengths and colors, jumper wires simplify circuit assembly and experimentation by providing a quick and efficient means of linking components without soldering. Widely used in projects ranging from hobbyist experimentation to professional development, these wires enhance the ease of circuit modification and troubleshooting.

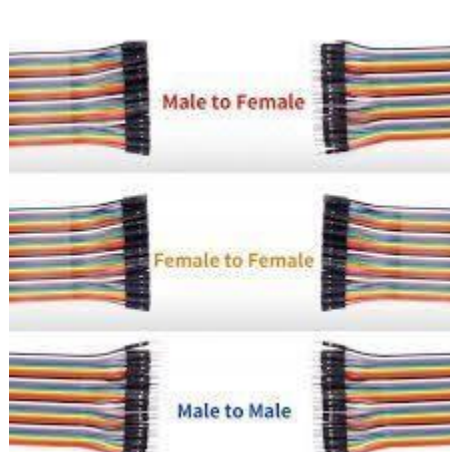


Fig -17: Jumper Wires

3.1.4 Hardware Connections:

The hardware implementation of the robot is successfully designed and it is given below.

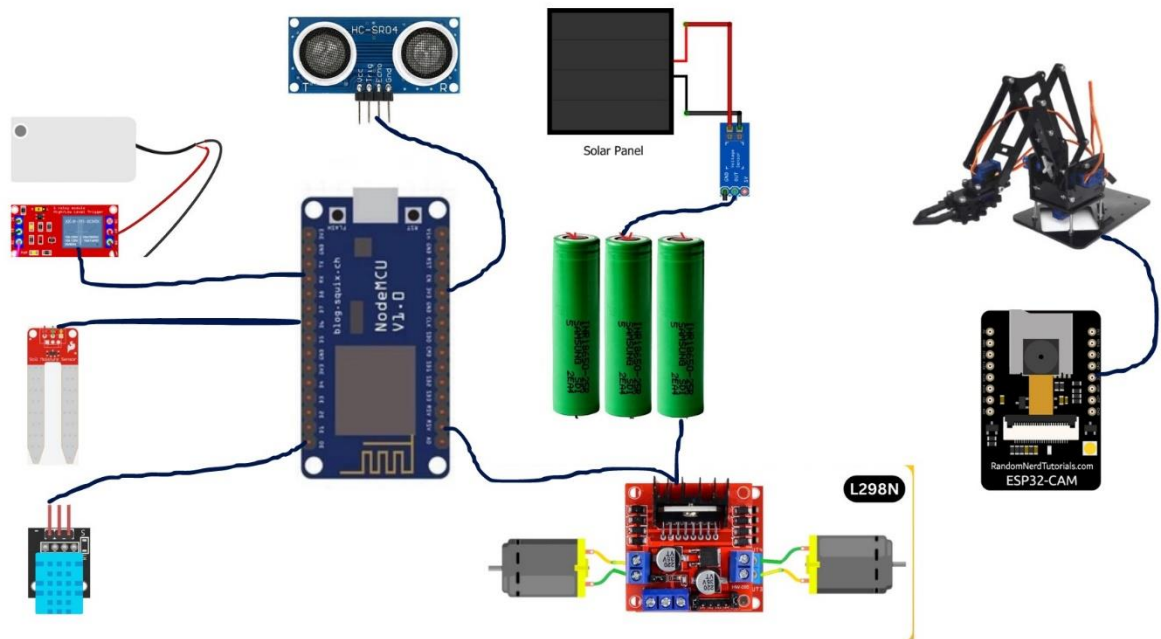
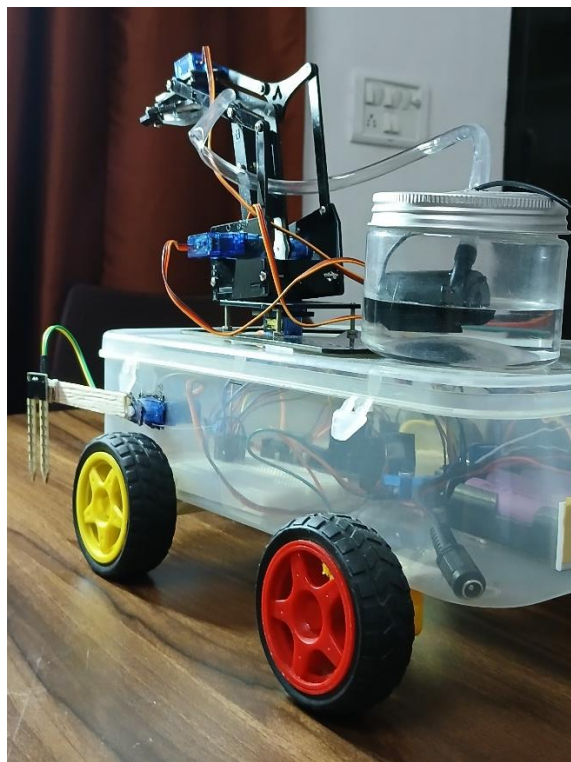


Fig -18: Block Diagram



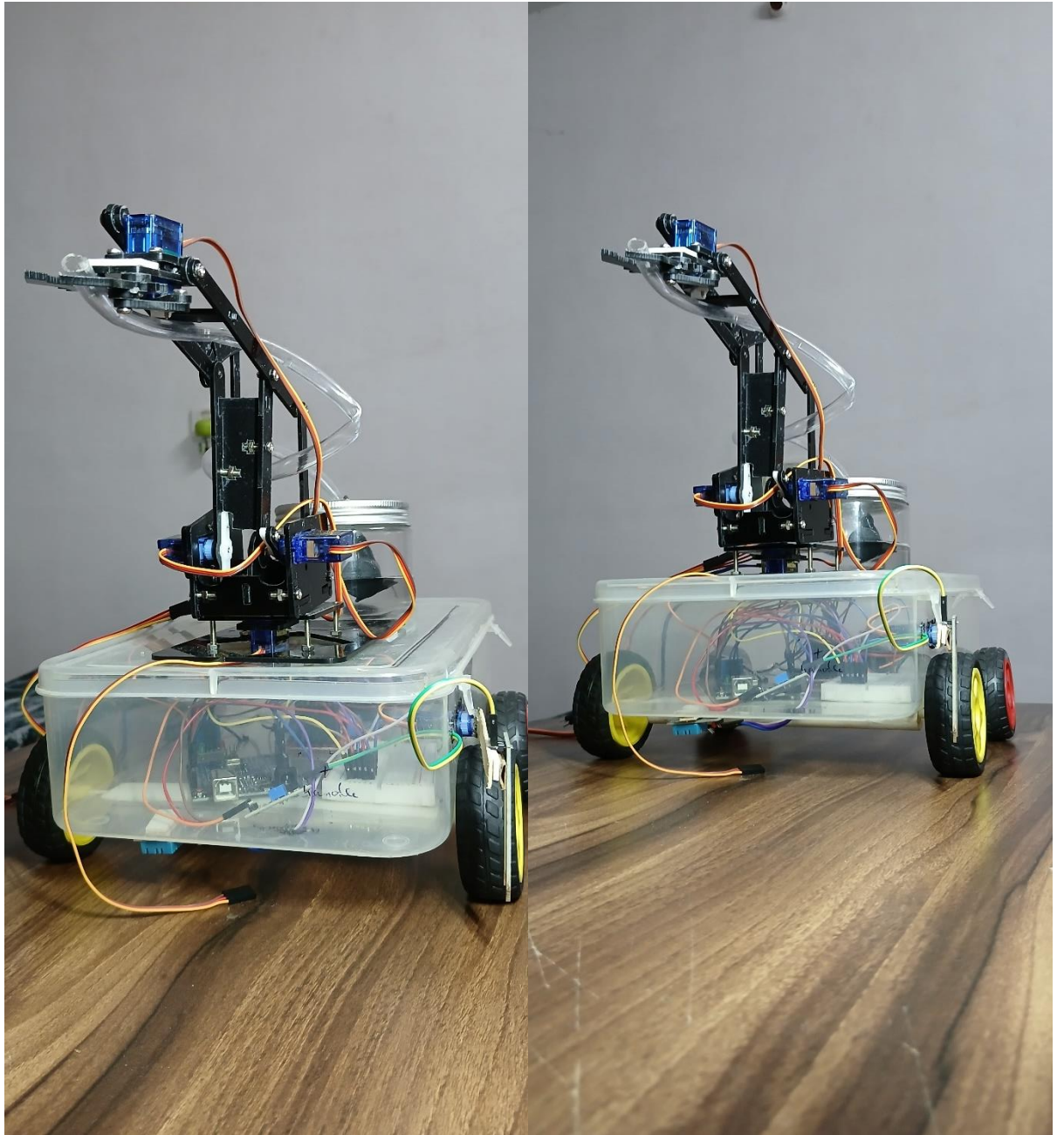


Fig -19: CropCrafter

3.2 Software Implementation

3.2.1 : Edge Impulse: Image Processing & ML Model Deployment

Edge Impulse is a platform that facilitates the development of machine learning models for edge devices, especially those used in IoT (Internet of Things) applications. It provides a comprehensive environment where developers can collect, label, preprocess, train, and deploy machine learning models directly onto embedded systems. The platform offers tools for data collection, signal processing, feature extraction, model training, and deployment optimization, all tailored for resource-constrained devices. This makes it easier for developers to implement AI functionalities directly on microcontrollers and other edge devices, without relying heavily on cloud services.

3.2.2 Steps for Implementation:

By following these steps, a robust object identification system using machine learning techniques can be developed and seamlessly integrated into the robotic arm's functionality.

1. **Data Collection:** Images of flowers, fruits, and orange bricks were gathered using an ESP32 device. Approximately 70 images were collected for each object.

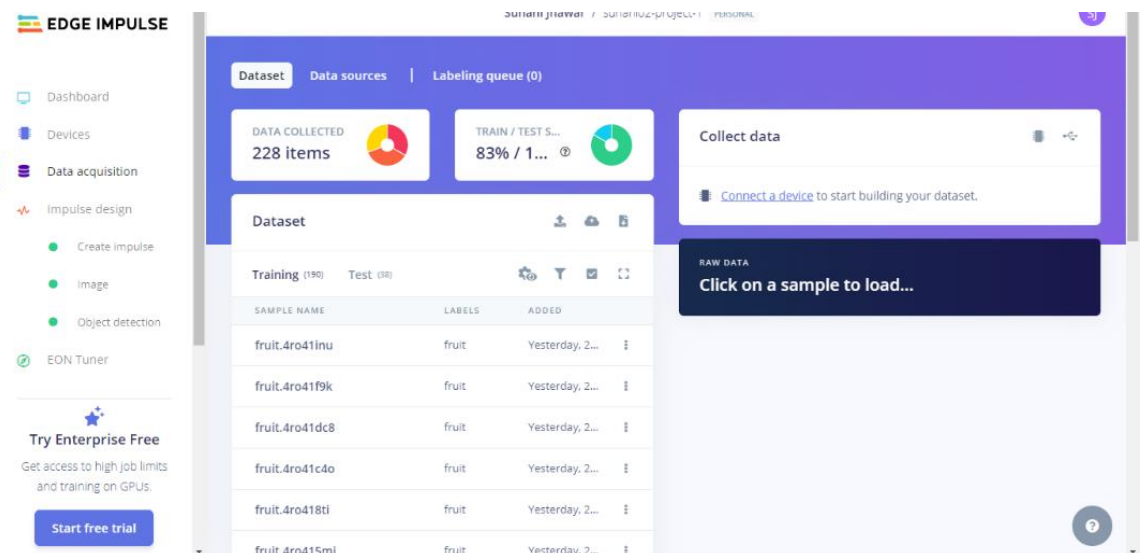


Fig -20: Data Collection- Edge Impulse

2. **Data Splitting:** The collected data was divided into training and testing sets, with 80% of the data allocated for training and the remaining 20% for testing.
- h
3. **Image Processing:** Image processing techniques were applied, converting the RGB images to

grayscale. This preprocessing step helps in reducing the computational complexity and extracting meaningful features.

4. **Feature Extraction and Selection:** Features were extracted from the preprocessed images. These features were then subjected to a selection process to identify the most relevant ones for the model.

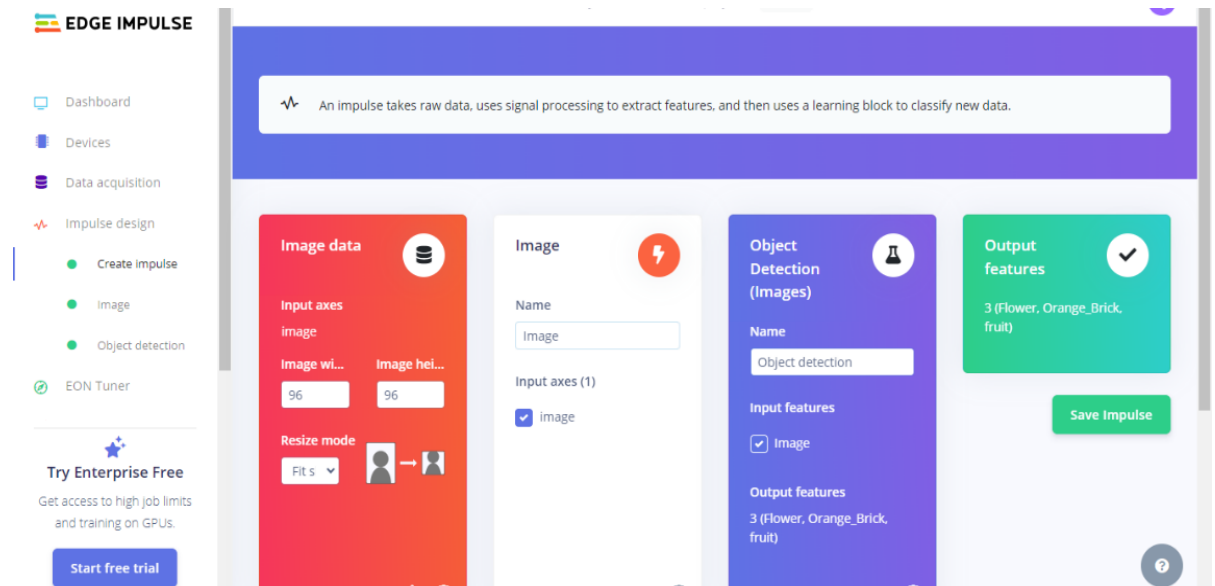
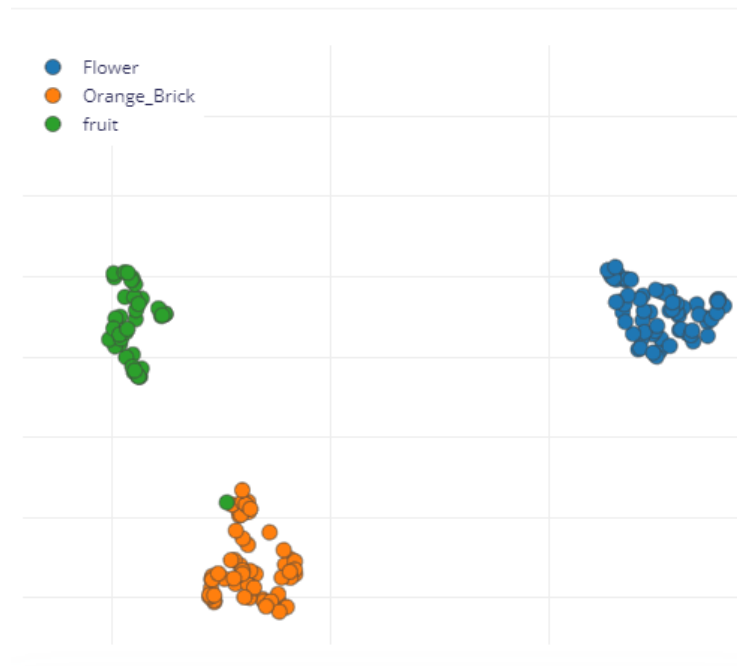


Fig -21: Feature Extraction- Edge Impulse

5. **Model Selection:** The FOMO (Faster Objects, More Objects) MobileNetV2 0.1 model was chosen for its efficiency and effectiveness in object recognition tasks.
6. **Model Training:** The selected model was trained using the training dataset, where it learned to distinguish between different objects based on the extracted features.
7. **Model Evaluation:** The trained model's performance was evaluated using the testing dataset. The accuracy achieved was 81.9%, indicating the model's ability to correctly classify objects.
8. **Confusion Matrix Analysis:** A confusion matrix was generated to further analyze the model's performance, providing insights into its ability to correctly classify different objects and any potential misclassifications.
9. **Deployment Preparation:** The trained model can be prepared for deployment as an Arduino library. This ensures compatibility and integration with the robotic arm system, allowing for real-time object identification and interaction.

Feature explorer



Model

Model version: ?

Quantized (int8) ▾

Last training performance (validation set)



F1 SCORE

81.9%

Confusion matrix (validation set)

	BACKGROUND	FLOWER	ORANGE_BRIC	FRUIT
BACKGROUND	99.8%	0.1%	0.1%	0.0%
FLOWER	20%	80%	0%	0%
ORANGE_BRIC	10%	0%	90%	0%
FRUIT	0%	0%	0%	100%
F1 SCORE	1.00	0.77	0.75	0.93

On-device performance ?

Engine: ?

EON™ Compiler ▾



INFERENCI...

1081
ms.



PEAK RAM ...

235.5K



FLASH US...

64.4K

Fig -22: Model Accuracy

3.2.3 Android Application:

The Crop Crafter project introduces a comprehensive mobile application designed to complement the functionality of the Crop Crafter robotic system. This intuitive app serves as a centralized hub for users to monitor and control various aspects of agricultural operations with ease and precision.

Key Features:

- **Sensor Measurements:** The app provides real-time data on soil moisture levels, environmental temperature, and humidity, enabling farmers to make informed decisions about irrigation timing and crop management.
- **Robotic Arm Control:** Users can remotely activate the gripper of the robotic arm through the app, facilitating precise manipulation of objects identified in the field.
- **Manual Irrigation:** In situations where manual intervention is necessary, users can initiate irrigation directly from the app, ensuring timely water supply to crops as needed.
- **Object Location:** The app displays the co-ordinates of object identified by the camera module, offering valuable insights for task prioritization and robotic arm maneuvering.

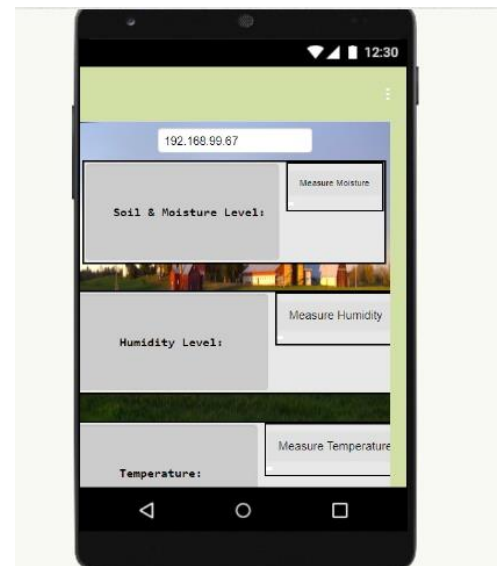
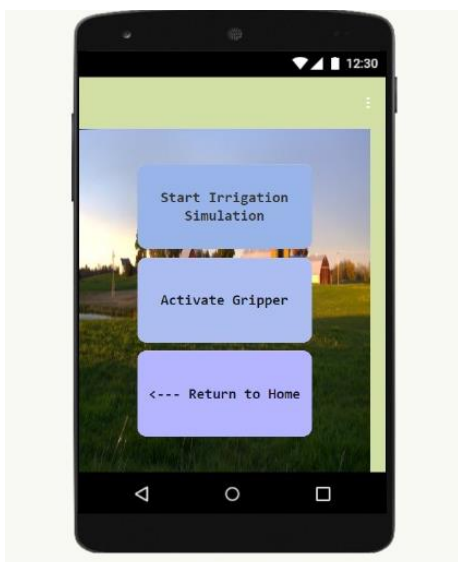


Fig -23: App. Logo

3.2.4 Application Screens and working behind it:



Fig -24 View of the Android App



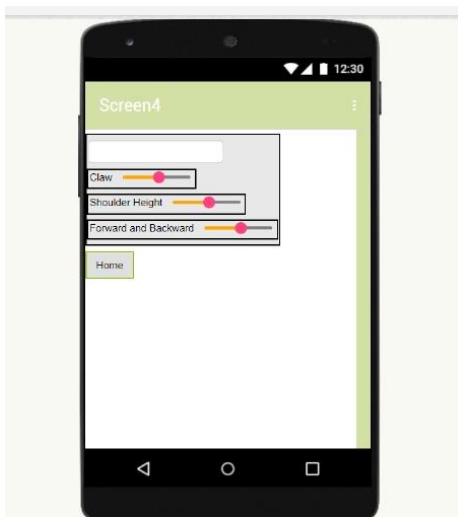


Fig-25 : App. Screens

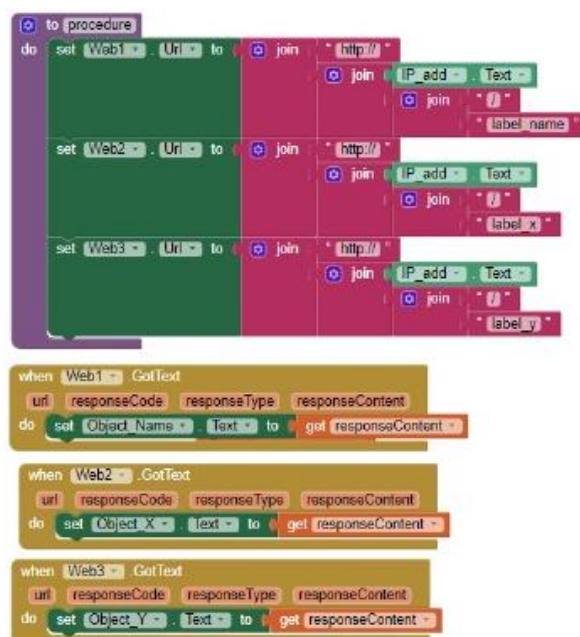


Fig-26 : Working of App.

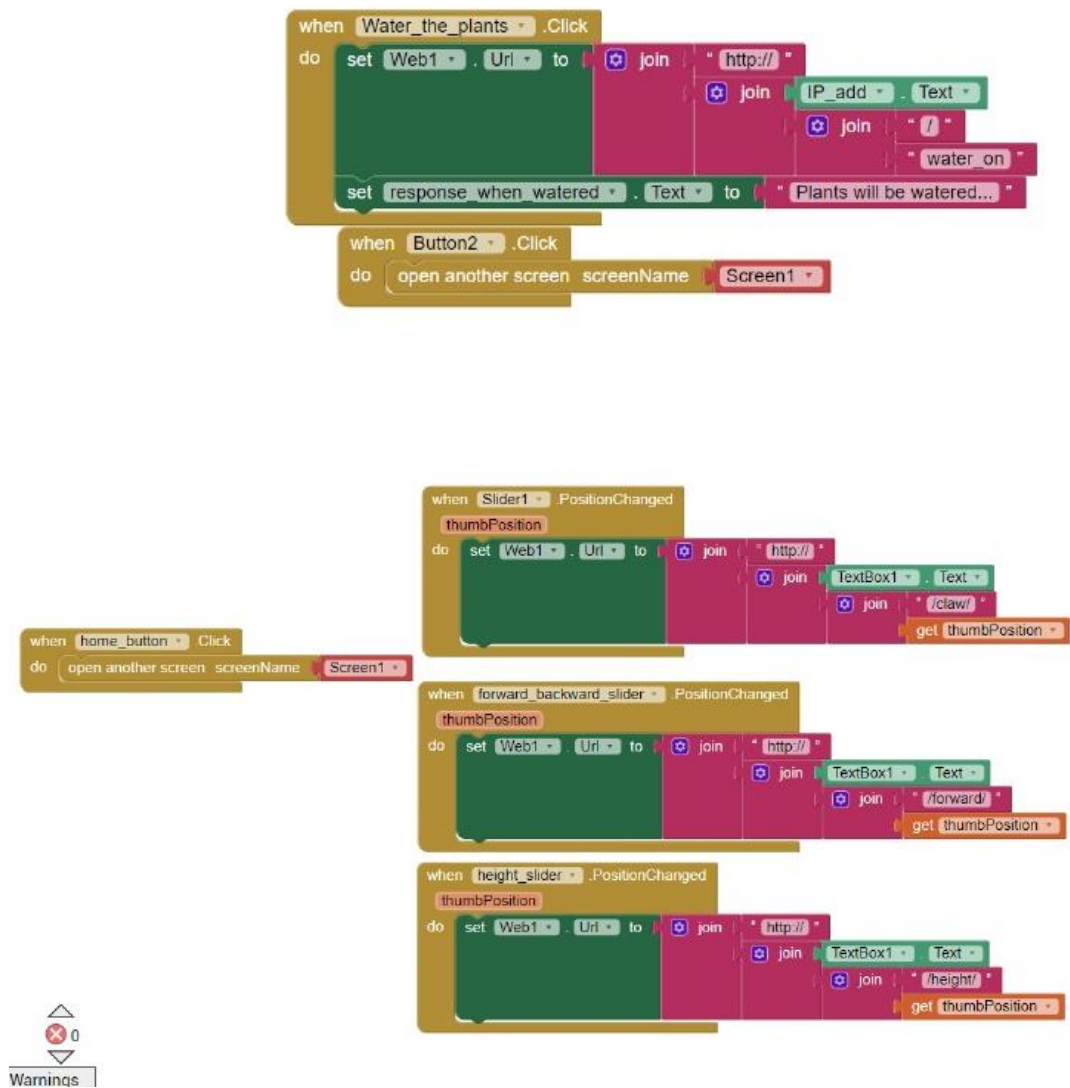


Fig-27 : Working of App for different command inputs.

3.2.5 Algorithm:

The algorithm behind the operation of the Crop Crafter robot is simple and understandable. Here's a breakdown of the process:

1. **Initialization:**

- Power on the CropCrafter agricultural robot.
- Initialize all hardware components, including ESP8266, ESP32-CAM, robotic arm, sensors, and actuators.
- Establish communication channels between ESP8266 and ESP32-CAM using ESP-NOW platform.
- Initialize the mobile app interface using MIT App Inventor and connect it to the local server.

2. **Sensor Data Collection:**

- Continuously monitor environmental conditions using sensors:
 - o DHT11 temperature and humidity sensor for ambient temperature and humidity.
 - o Moisture sensor for soil moisture level.
 - o Ultrasonic sensors for obstacle detection and avoidance.
- Read sensor data periodically at predefined intervals.

3. **Image Processing and Object Detection:**

- Capture images using the ESP32-CAM camera module.
- Perform machine learning-based object detection and classification using the Edge Impulse platform:
 - o Load the pre-trained machine learning model into ESP32-CAM.
 - o Capture images of crops and other objects in the field.
 - o Process images using the machine learning model to identify and classify objects.
 - o Determine the location and characteristics of detected objects.

4. **Robotic Arm Control:**

- Receive object detection results from ESP32-CAM via ESP-NOW communication.
- Analyze object detection data to determine the location and type of detected objects.
- Calculate the coordinates and trajectory for the robotic arm to reach and manipulate the detected objects.
- Control the robotic arm actuators to move the arm to the desired position and perform tasks such as picking fruits or adjusting sensors.

5. **Irrigation Management:**

- Monitor soil moisture levels using the moisture sensor.
- Analyze soil moisture data to determine the need for irrigation.

- Activate the water pump and open/close the water valve based on irrigation requirements:
 - Activate the water pump to draw water from the water source (e.g., tank or reservoir).
 - Open the water valve to irrigate the soil in areas with low moisture levels.
 - Monitor irrigation duration and volume to prevent overwatering.

6. Obstacle Avoidance:

- Continuously scan the surroundings using ultrasonic sensors for obstacles.
- Detect obstacles in the path of the robot and calculate avoidance maneuvers:
 - Adjust the robot's trajectory to navigate around obstacles.
 - Stop or slow down the robot's movement if necessary to avoid collisions.
 - Update the navigation path dynamically based on real-time obstacle detection.

7. User Interaction:

- Provide real-time feedback and status updates to the user via the mobile app interface:
 - Display environmental data such as temperature, humidity, and soil moisture levels.
 - Show object detection results and robotic arm actions.
 - Enable user control and interaction with the robot, such as initiating tasks or adjusting settings.

8. Error Handling and Recovery:

- Implement error detection and recovery mechanisms to handle unexpected situations:
 - Monitor system health and detect hardware or software failures.
 - Log errors and exceptions for diagnostic purposes.
 - Attempt to recover from errors autonomously or prompt user intervention if necessary.

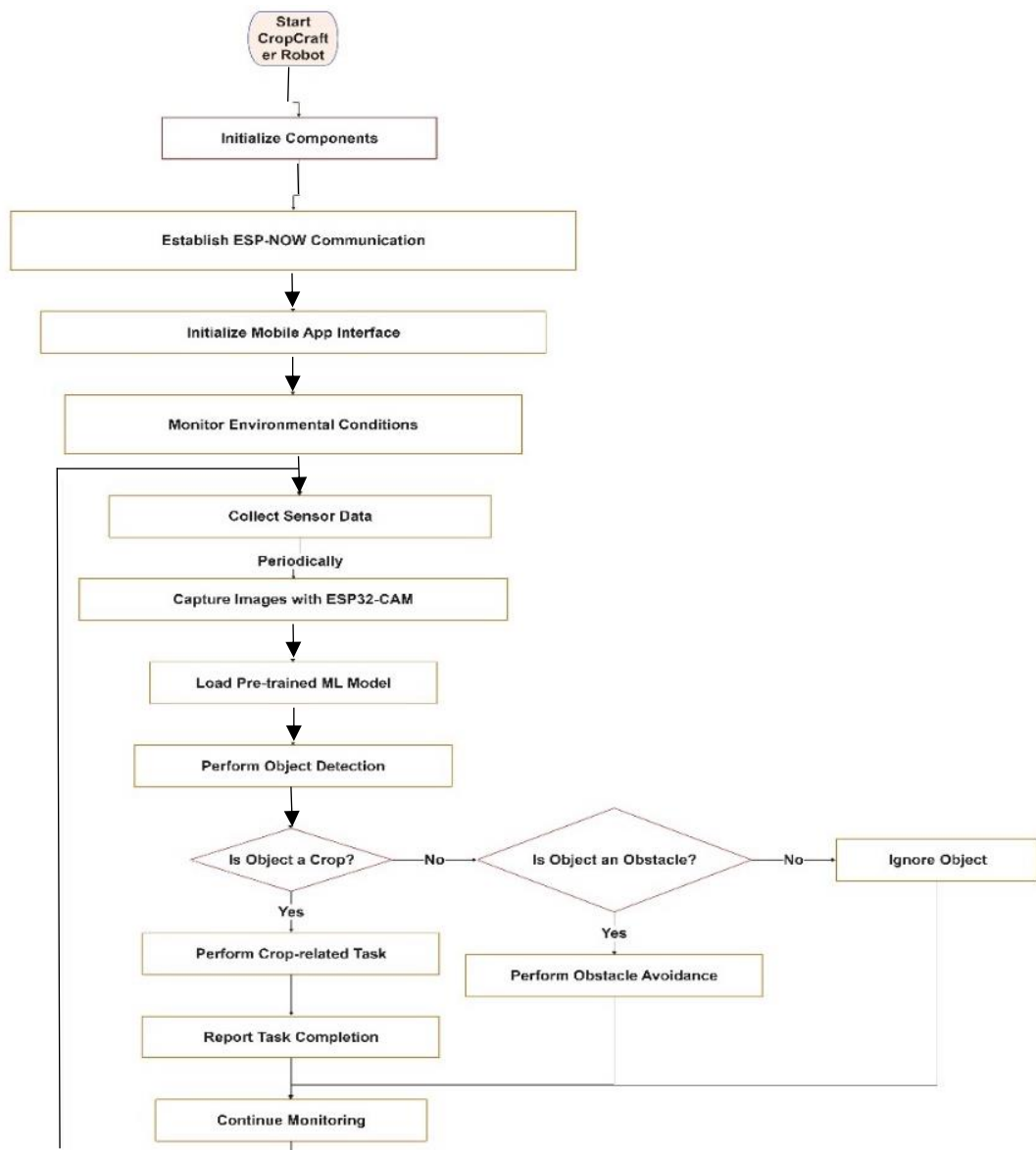


Fig -28: Algorithm

Chapter 4: Final Analysis and Review

Crop Crafter represents a major change in farming, making things much easier by automating tasks like watering, checking soil moisture, identifying seeds using image processing and machine learning, and harvesting, with the added capability of a robotic arm for object manipulation. This means less need for people to do these jobs manually. With everything working together smoothly, it helps farms produce more efficiently, using resources wisely and managing crops better. It's a big step forward in farming tech, complemented by its accompanying mobile app.

4.1 Challenges:

1. Initial Costs and Maintenance: The initial investment for Crop Crafter is substantial, encompassing high-tech components and assembly. Ongoing maintenance adds to expenses, requiring regular servicing and parts replacement.
2. Complex Construction: The Crop Crafter's intricate design poses challenges in assembly and integration. Coordinating various components demands planning and technical expertise.
3. Seed Identification: Utilizing Edge Impulse for seed identification requires extensive image collection for accurate machine learning, which is time-consuming and labor-intensive.

4.2 Conclusion:

In conclusion, Crop Crafter heralds a transformative era in modern agriculture, spearheading a revolution in farming methodologies with its cutting-edge technology. This pioneering ag robot seamlessly integrates advanced features to optimize crop cultivation and management, empowering farmers to boost productivity while curbing expenses. Leveraging sophisticated image processing capabilities, it precisely identifies crop types in the field, tailoring management strategies accordingly. Equipped with sensors to monitor soil conditions in real-time, it ensures optimal growing environments and minimizes water usage through precision irrigation systems. Navigating safely with ultrasonic sensors, its versatile robotic arm undertakes various tasks from harvesting to seed sowing with unparalleled efficiency. Operating sustainably on rechargeable batteries and solar panels, Crop Crafter exemplifies the future of precision farming.

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